

Tagged Restricted Pro-Ab Realization

Date: 2026-06-11

Summary

Pass 74 tests the first concrete external realization of $\mathcal{H}_\epsilon$.

The target is the tagged restricted pro-Ab certificate category

$\mathbf{Pro}^{\mathrm{rp}}_{\mathrm{tag}}(\mathbf{Ab}^{\mathrm{fin}}) \times \mathbf{Pro}_{\mathrm{tag}}(\mathbf{Ab}^{\mathrm{fin}})$

This is not yet an LCA-sheaf or condensed realization. It is a concrete finite/pro target used to test faithfulness on the five generator families.

Realization Functor

The functor

$$\rho_{\mathrm{tag}} : \mathcal{H}_\epsilon \rightarrow \mathcal{R}_\epsilon$$

sends:

- finite conductor windows to tagged finite abelian group presentations with support tag S , conductor tag k , elementary divisors $(p, 2k)$, and lattice divisors (p, k) ;
- Loeb-Rosser boundaries d_S to tagged integer matrices;
- restrictions $S \subseteq S'$ to tagged coordinate-restriction matrices;
- signed duality to $-d_S^T$;
- lcm kernel stages $K_n = N_n \mathbb{Z}$ to tagged pro-stages (n, N_n) .

Result

On the checked finite/pro window, ρ_{tag} is faithful on all five generator families:

1. finite conductor windows;
2. Loeb-Rosser boundaries;
3. restrictions;
4. signed duality;
5. derived pro-Ab lcm stages.

The corresponding plain tag-forgetting target is not faithful. Restriction source supports collide after forgetting source tags, and repeated lcm stages collide after forgetting stage tags.

Verification

The checker `code/scripts/check-pass74.py` produces `artifacts/reports/pass74-tagged-proab-realization-check.json`.

It verifies:

- 75 generators through six primes, conductors $k \leq 3$, and lcm stages through N_{24} ;
- zero collisions for tagged signatures;
- family faithfulness for all five generator families;
- collisions in the plain tag-forgetting comparison;

- signed-dual double-dual compatibility.

The remaining problem is to make the tags intrinsic: support and stage data should become genuine structure in an LCA-sheaf, condensed/solid, or canonical exact pro-category target, or else a no-go theorem should explain why this cannot be done.